Notes: Ball & Mankiw (RES, 2023)
Market Power in Neoclassical Growth Models

Contents

1 Big picture

- **Question.** What happens to standard neoclassical growth and public-debt results when firms have market power and charge prices above marginal cost?
- **Core mechanism.** With market power, firms choose capital so that the marginal product of capital exceeds the rental cost of capital. Therefore, the real interest rate earned by savers can be below the social return on capital.
- **Main implication.** The real interest rate alone is no longer a sufficient statistic for dynamic efficiency.
- **Calibration result.** With firm markup $\gamma = 1.2$ and Tobin's $q = 1.75$, the calibration implies $r = 4.3\%$, measured return $m = 6.0\%$, and social return $f'(k) - \delta = 8.3\%$.
- **Fiscal-policy message.** Even if $r < g$ makes a Ponzi scheme feasible, debt can reduce welfare if the social return on capital exceeds g.

2 Incremental contribution of the paper

- The paper adds market power to the Solow and Diamond models.
- It shows that dynamic efficiency should be evaluated using the social return on capital, not only r.
- It separates public-debt sustainability from welfare desirability.
- It shows that feasible Ponzi schemes and bubbles may be welfare-reducing in a dynamically efficient economy with market power.
- It provides a calibration showing that the private-social return wedge can be several percentage points.

3 Model objects and measurement

3.1 Markup wedge

Firms with market power set price above marginal cost. The rental price of capital satisfies

$$R = \frac{MPK}{\mu}, \qquad r = \frac{MPK}{\mu} - \delta.$$

Interpretation: The social return is $MPK - \delta$, while savers earn r. When $\mu > 1$, these differ.

3.2 Calibration objects

$$\mu = \frac{(1-\lambda)\gamma}{1-\lambda\gamma}, \qquad m = \frac{1-\phi}{k/y} - \delta, \qquad r = \frac{f'}{\mu} - \delta.$$

The baseline uses $\gamma = 1.2$, $q = 1.75$, and $\lambda = 0.45$.

4 Models and theoretical specifications

4.1 Solow model

Steady-state consumption per effective worker is

$$c = f(k) - (\delta + g)k,$$

and the golden-rule condition is

$$f'(k_{GR}) - \delta = g.$$

Interpretation: The efficient capital stock depends on the social marginal product, not on the private real interest rate.

4.2 OLG model and debt

Young agents save for old-age consumption:

$$C_{y,t} = F[K_t, A_t(1-X)] - R_t K_t + \tau_t - S_t, \qquad C_{o,t+1} = R_{t+1} S_t.$$

The Euler equation is

$$\frac{C_{o,t+1}}{C_{y,t}} = \beta R_{t+1}.$$

If debt is kept at $B_t = A_t b$, transfers per effective worker satisfy

$$\frac{\tau_t}{A_t} = \frac{g-r}{1+g} b.$$

Interpretation: If $r < g$, debt can be rolled over with positive transfers. But this does not imply welfare gains when market power depresses r.

4.3 Debt welfare decomposition

Debt has two effects:

- **Aggregate effect:** debt crowds out capital.
- **Generational effect:** debt reallocates consumption between young and old.

If $r < g < f'(k) - 1$, these two effects conflict.

5 Identification and theoretical design

The paper is theoretical and calibration-based. The wedge between r and the social return follows mechanically from markups. The calibration asks whether plausible markups and q values make the wedge large enough to change conclusions about dynamic efficiency and debt policy.

6 Tables and theoretical result blocks

6.1 Table 1: Sensitivity check

Read: Table 1 shows how changing γ and q changes r, measured return m, social return $f' - \delta$, pure profits, and overhead labour.

Detailed interpretation: In the baseline, $r = 4.3\%$, $m = 6.0\%$, and $f' - \delta = 8.3\%$. The social return remains far above r under plausible market-power calibrations.

Economic message: Low private returns need not imply low social returns.

TABLE 1

Sensitivity check

	r	m	$f'-\delta$	Π	NX/L
Competitive benchmark					
$\gamma=1.0, q=1.0$	6.0	6.0	6.0	0	0
Effect of varying γ (with $q=1.75$)					
$\gamma=1.05$	4.3	6.0	5.2	5.1	5
$\gamma=1.1$	4.3	6.0	6.1	5.1	17
$\gamma=1.2$ (baseline)	4.3	6.0	8.3	5.1	38
$\gamma=1.3$	4.3	6.0	11.0	5.1	55
$\gamma=1.4$	4.3	6.0	14.3	5.1	70
Effect of varying q (with $\gamma=1.2$)					
$q=1.0$	6.0	6.0	10.8	0	45
$q=1.25$	5.2	6.0	9.6	2.4	42
$q=1.5$	4.7	6.0	8.9	4.0	39
$q=1.75$ (baseline)	4.3	6.0	8.3	5.1	38
$q=2.0$	4.0	6.0	7.9	6.0	36
$q=2.25$	3.8	6.0	7.6	6.7	35

Notes: All entries are in percentage points.

Table 1: Sensitivity check

TABLE 2

Sensitivity check, continued

The social return on capital: $f'-\delta$						
	$\gamma=1.0$	$\gamma=1.05$	$\gamma=1.1$	$\gamma=1.2$	$\gamma=1.3$	$\gamma=1.4$
$q=1.0$	6.0	7.0	8.2	10.8	14.0	17.9
$q=1.25$	-	6.2	7.2	9.6	12.6	16.2
$q=1.5$	-	5.6	6.6	8.9	11.7	15.1
$q=1.75$	-	5.2	6.1	8.3	11.0	14.3
$q=2.0$	-	4.9	5.8	7.9	10.5	13.7
$q=2.25$	-	4.6	5.5	7.6	10.1	13.3
The wedge between the social return on capital and the real interest rate: $f'-\delta-r$						
	$\gamma=1.0$	$\gamma=1.05$	$\gamma=1.1$	$\gamma=1.2$	$\gamma=1.3$	$\gamma=1.4$
$q=1.0$	0.0	1.0	2.2	4.8	8.0	11.9
$q=1.25$	-	1.0	2.0	4.4	7.4	11.0
$q=1.5$	-	0.9	1.9	4.2	7.0	10.5
$q=1.75$	-	0.9	1.8	4.0	6.7	10.0
$q=2.0$	-	0.9	1.8	3.9	6.5	9.7
$q=2.25$	-	0.8	1.7	3.8	6.3	9.5

Notes: All entries are in percentage points.

Table 2: Sensitivity check, continued

6.2 Table 2: Sensitivity check, continued

Read: Table 2 varies q and γ jointly. The first panel reports $f' - \delta$; the second panel reports $f' - \delta - r$.

Detailed interpretation: The wedge is large whenever market power is significant. Markup variation matters more than q variation for the size of the wedge.

Economic message: Market power makes the private-social return wedge robust and quantitatively important.

6.3 Result block: Dynamic efficiency

With competition, r equals the social return. With market power, $r = f'(k)/\mu - \delta$, while the social return is $f'(k) - \delta$. Therefore $r < g$ does not imply dynamic inefficiency.

6.4 Result block: Public debt

If $r < g$, debt rollover is feasible. But if $f'(k) - 1 > g$, debt crowds out capital with a social return above growth and may lower welfare.

7 Short summary

- Market power lowers the private return on capital below the social return.
- The real interest rate is not a sufficient statistic for dynamic efficiency.
- A plausible calibration implies a social capital return around 8.3%, well above growth.
- Government debt may be sustainable when $r < g$ but welfare-reducing if it crowds out productive capital.
- The welfare effect of debt combines an aggregate effect and a generational effect.

8 Expanded conclusion

8.1 Core conclusion

Market power changes the interpretation of low real interest rates. Low r does not necessarily mean capital has low social value.

8.2 Economic intuition

Firms with markups invest until the private return equals their cost, but the social marginal product remains higher. The wedge can be large enough to affect fiscal-policy conclusions.

8.3 Fiscal implication

Debt can be feasible and still harmful. Sustainability is not the same as desirability.

8.4 Limitations

The paper abstracts from uncertainty, risk premia, heterogeneous households, and detailed fiscal instruments. It is a conceptual and calibration exercise.

8.5 Takeaway

Fiscal policy should compare growth to the social return on capital, not only to the government borrowing rate.